

Good morning, Professor and peers. Today, I am presenting our implementation of a classical Computer Vision pipeline for Medical Image Morphological Processing on Brain MRI scans.

In medical imaging, machine learning models often act as black boxes. Before deploying complex neural networks, classical mathematical morphology provides deterministic, interpretable, and computationally lightweight segmentation. Our objective is to isolate anatomical structures, filter background artifacts, and extract precise boundaries using standard morphological operations in OpenCV.

Slide 1: Data Acquisition & Preprocessing

- **Visual on Screen:** Grayscale Brain MRI scan (256×256).
- **Spoken Script:**
"Starting with Step 1: We ingest the raw axial brain MRI scan. The image is loaded directly in single-channel grayscale using OpenCV and normalized to a standard spatial resolution of 256×256 pixels. Working in grayscale reduces computational overhead from 3D RGB tensors to a 2D intensity matrix $I(x, y)$, where pixel values strictly represent tissue radiodensity."

Slide 2: Statistical Profiling & Intensity Distribution

- **Visual on Screen:** Intensity Histogram with Mean ($\mu = 119.79$), Standard Deviation ($\sigma = 49.75$), Range $[12, 255]$.
- **Spoken Script:**
"In Step 2, we perform exploratory intensity analysis. The pixel distribution shows a mean intensity of approximately 119.8 and a standard deviation of 49.75 , indicating moderate contrast across white matter, gray matter, and cerebrospinal fluid (CSF). Notice the histogram: the sharp peak near zero represents the dark background air void, while the broad distribution between 80 and 220 represents cranial tissue. This bimodal characteristic makes the image an ideal candidate for global thresholding."

Slide 3: Binarization via Otsu's Global Thresholding

- **Visual on Screen:** Binary Mask with $T^* = 95$.
- **Spoken Script:**
"Step 3 applies Otsu's Automatic Thresholding. Rather than picking an arbitrary manual cutoff, Otsu's algorithm minimizes the intra-class variance—or equivalently, maximizes inter-class variance—between background and foreground. The calculated optimal threshold T^ is 95 . Every pixel $I(x, y) \geq 95$ becomes 1 (foreground brain tissue), and $I(x, y) < 95$ becomes 0 (background void). This yields a clean binary mask B ."*

Slide 4: Mathematical Morphology — Core Operations

- **Visual on Screen:** Side-by-side comparison: Structuring Element (B), Erosion, Dilation, Opening, Closing.
- **Spoken Script:**
"With our binary mask established, we apply mathematical morphology using a defined structuring element (kernel) S :
 - **Erosion ($A \ominus S$):** Shrinks the foreground, stripping away thin cranial noise and isolated artifact pixels.
 - **Dilation ($A \oplus S$):** Expands boundaries, bridging micro-fissures and restoring tissue mass.
 - **Opening ($A \circ S = (A \ominus S) \oplus S$):** Erosion followed by dilation—eliminates small background noise while preserving overall shape geometry.
 - **Closing ($A \bullet S = (A \oplus S) \ominus S$):** Dilation followed by erosion—fills internal holes and smooths contours without altering outer dimensions."

Slide 5: Boundary & Feature Extraction

- **Visual on Screen:** Morphological Gradient / Boundary Extraction result.
- **Spoken Script:**
"Finally, we perform Morphological Gradient Boundary Extraction. By computing the set difference between the dilated mask and the eroded mask $\beta(A) = (A \oplus S) - (A \ominus S)$, we isolate the exact perimeter of the brain parenchyma. This boundary mask serves as an anatomical region of interest (ROI) for downstream tumor or lesion localization."

To conclude: we constructed an end-to-end, deterministic computer vision pipeline without relying on trained weights or GPU overhead. By combining statistical thresholding with morphological filtering and boundary extraction, we achieved robust anatomical segmentation. Thank you, and I am now open to your questions.

Likely Question	Model Technical Answer
Why use Otsu over Adaptive Thresholding?	<i>"Adaptive thresholding computes thresholds locally, which is great for uneven illumination (like shadows), but MRI scans have relatively uniform field bias. Global Otsu thresholding cleanly separates the bimodal distribution of cranial tissue versus background air."</i>

Why not use Deep Learning (e.g., U-Net)?	<i>"Deep learning requires labeled masks, compute resources, and runs risk of hallucination/overfitting on small datasets. Morphological operations provide a fast, deterministic baseline that works on single zero-shot images."</i>
What determines the choice of Structuring Element (Kernel)?	<i>"A standard 3×3 or 5×5 elliptical/rectangular kernel reflects natural anatomical curvature. A kernel too large causes excessive structural erosion, destroying delicate cortical sulci."</i>